Phase 5

Evaluation Report CardKnowlogy Expert System

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The final evaluation of the **CardKnowlogy**. The primary focus is to test the system against realistic scenarios, evaluate the correctness and consistency of its outputs, and analyze the logic used to reach clinical decisions in a telemedicine environment.

1. Evaluation Objectives

The evaluation aimed to verify that the **Rule-Based Expert System (RBES)** could effectively act as an intelligent assistant for early triage. Key objectives included:

• **Correctness of Outputs**: Ensuring the system identifies conditions accurately based on clinical thresholds (e.g., Oxygen Saturation < 90% or Systolic BP ≥ 140).

• **Consistency of Reasoning**: Confirming that the LEX strategy and Certainty Factor (CF) combination formula produce stable, deterministic results.

• **Transparency**: Demonstrating the system's ability to explain "why" a conclusion was reached using an explicit Audit Trail.

• **Safety Evaluation**: Assessing how the system handles incomplete data to prevent unsafe triage guesses.

2. Manual Testing Evaluation

Case 1 :Critical Emergency

-Manual solution

Initial Facts (Working Memory):
• **F1:** Heart disease (CF=0.85)
• **F2:** Syncope (CF=0.95)
• **F3:** Oxygen Saturation < 85 (CF=0.98) — **Newest**

Execution Trace:

Cycle 1

- **Conflict Set:** {R56 (F2, F3)}
- **Resolution (LEX): R56 (F2, F3)** wins (matches newest fact **F3**).
- **Calculation:** Min(F2, F3) = 0.95. CF_{R56} = 0.98 * 0.95 = 0.931.
- **Action:** Add **F4: Disease = Sudden Cardiac Arrest** (CF=0.931).

Cycle 2

- **Conflict Set:** {R61 (F4), R62 (F4)}
- **Resolution (LEX): R61 (F4)** wins (Recency: matches newest fact **F4**; Order: R61 comes before R62).
- **Action:** Add **F5: Urgency = CRITICAL**.

Cycle 3

- **Conflict Set:** {R62 (F4)}
- **Resolution (LEX): R62 (F4)** wins.
- **Action:** Add **F6: Recommendation exists**.

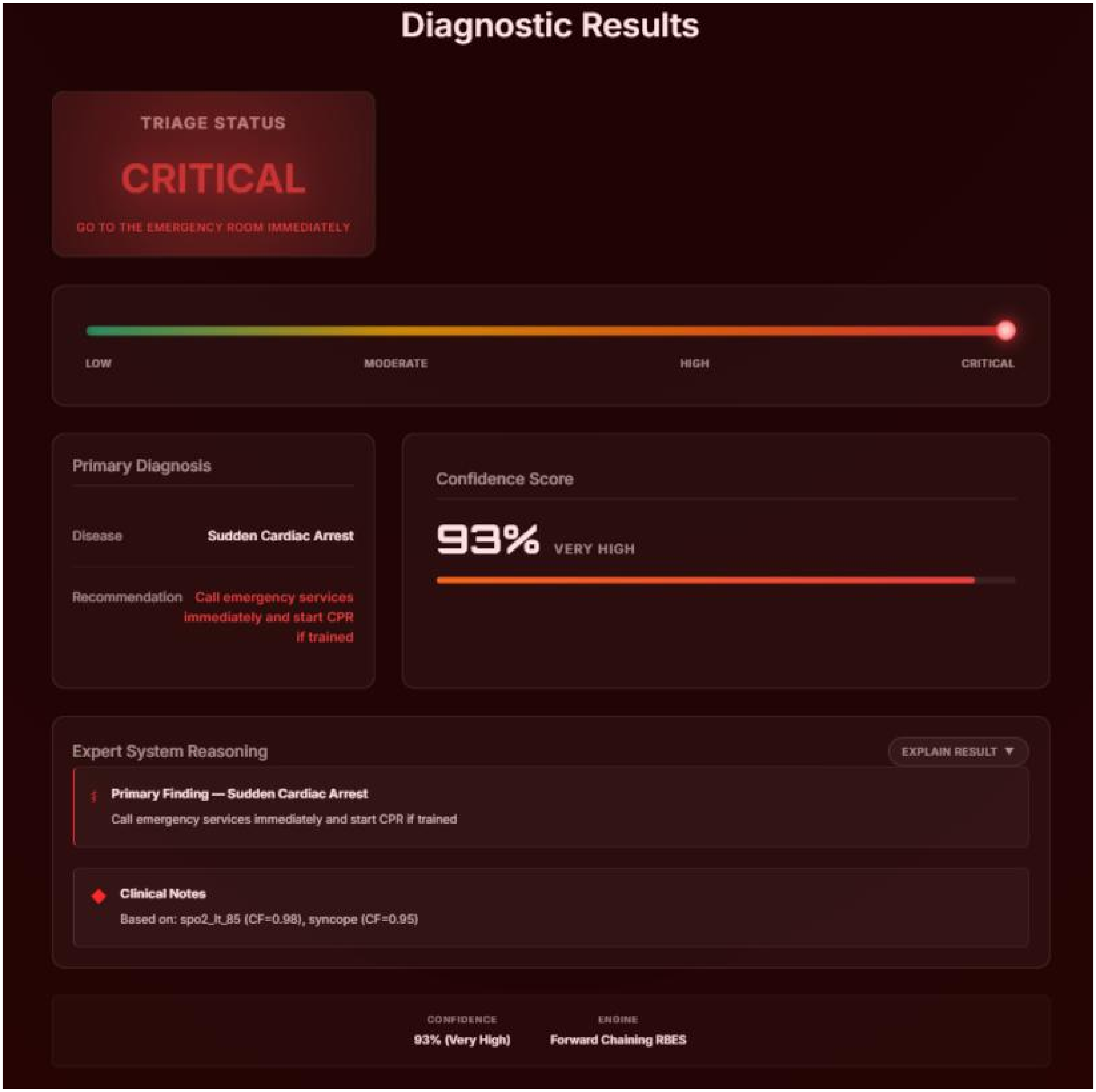
Cycle 4

- **Conflict Set:** {R0 (F4, F5, F6)}
- **Resolution (LEX): R0 (F4, F5, F6)** wins (matches latest fact **F6**).
- **Action:** Print Output. **Final Output:**
- **Disease:** Sudden Cardiac Arrest
- **Urgency:** CRITICAL
- **Advice:** Call emergency services immediately and start CPR if trained

Final Output:

- **Disease:** Sudden Cardiac Arrest
- **Urgency:** CRITICAL
- **Advice:** Call emergency services immediately and start CPR if trained

-The system output



Case 2: Low Urgency

-Manual solution

Initial Facts (Working Memory):

- **F1:** Heart disease (CF=0.85)
- **F2:** Chest pain (CF=0.90)
- **F3:** Low activity (CF=0.55) — **Newest**

Execution Trace:

Cycle 1

- **Conflict Set:** {R41 (F1, F2, F3)}
- **Resolution (LEX): R41 (F1, F2, F3)** wins (matches newest fact **F3**).
- **Calculation:** Min(F1, F2, F3) = 0.55 CF_{R41} = 0.80 * 0.55 = 0.44.
- **Action:** Add **F4: Disease = Stable Angina** (CF=0.44).

Cycle 2

- **Conflict Set:** {R43 (F4), R44 (F4)}
- **Resolution (LEX): R43 (F4)** wins (matches newest fact **F4**).
- **Action:** Add **F5: Urgency = LOW**.

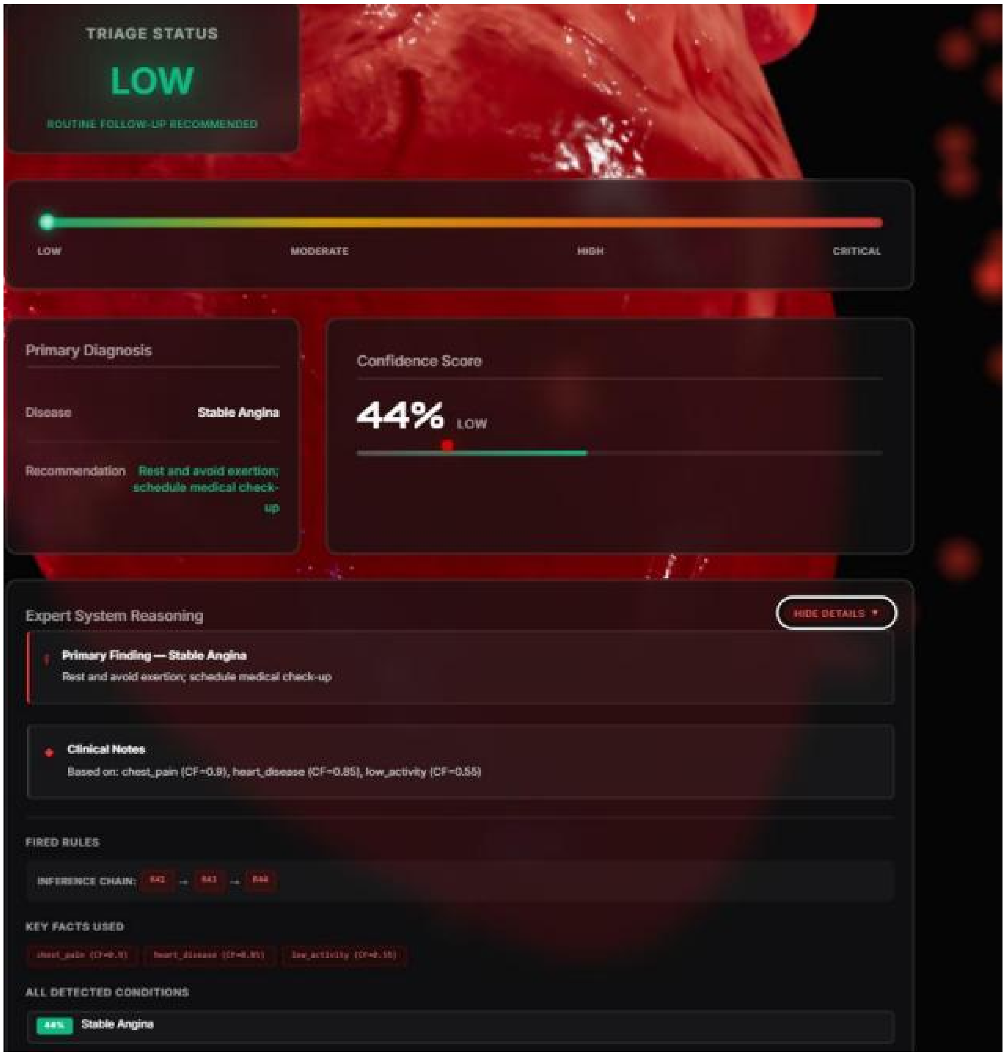
Cycle 3

- **Conflict Set:** {R44 (F4)}
- **Resolution (LEX): R44 (F4)** wins.
- **Action:** Add **F6: Recommendation exists**.

Final Output:

- **Disease:** Stable Angina
- **Urgency:** LOW
- **Advice:** Rest and avoid exertion; schedule medical check-up

-The system output



Case 3: Multiple Overlapping Conditions

-Manual solution

Initial Facts (Working Memory):

- **F1:** Chest pain (CF=0.90)
- **F2:** Shortness of breath (CF=0.80)
- **F3:** Edema (CF=0.75)
- **F4:** Diabetes (CF=0.70) — **Newest**

Execution Trace:

Cycle 1

- **Conflict Set:** {R35 (F1, F2), R37 (F1, F4)}
- **Resolution (LEX): R37 (F1, F4)** wins (matches newest fact **F4**).
- **Calculation:** Min(F1, F4) = 0.70 CF_{R37} = 0.78* 0.70 = 0.546
- **Action:** Add **F5: Disease = Acute Coronary Syndrome (ACS)** to PCS.

Cycle 2

- **Conflict Set:** {R35 (F1, F2), R39 (F5), R40 (F5)}
- **Resolution (LEX): R35 (F1, F2)** wins (matches oldest facts but is a primary diagnosis rule).
- **Calculation:** Min(F1, F2) = 0.80 CF_{R35} = 0.85* 0.80 = 0.68
- **Action: Evidence Aggregation** for ACS.
- **Combining CFs:** CF_{new} = 0.546 + 0.68 - (0.546 * 0.68) = 0.8547
- **Action:** Update **F5: Disease = ACS** with CF = 0.855.

Cycle 3

- **Conflict Set:** {R69 (F1, F2, F3), R39 (F5), R40 (F5)}
- **Resolution (LEX): R69 (F1, F2, F3)** wins (Specific condition: 3 inputs).
- **Calculation:** Min(F1, F2, F3) = 0.75 CF_{R69} = 0.80 * 0.75 = 0.60
- **Action:** Add **F6: Disease = Acute Decompensated Heart Failure (ADHF)** to PCS.

Cycle 4 (Primary Selection)

- **PCS Ranking:** 1. **ACS:** CF = 0.855 2. **ADHF:** CF = 0.60
- **Resolution: ACS** is selected as the **Primary Condition** because it has the highest CF.

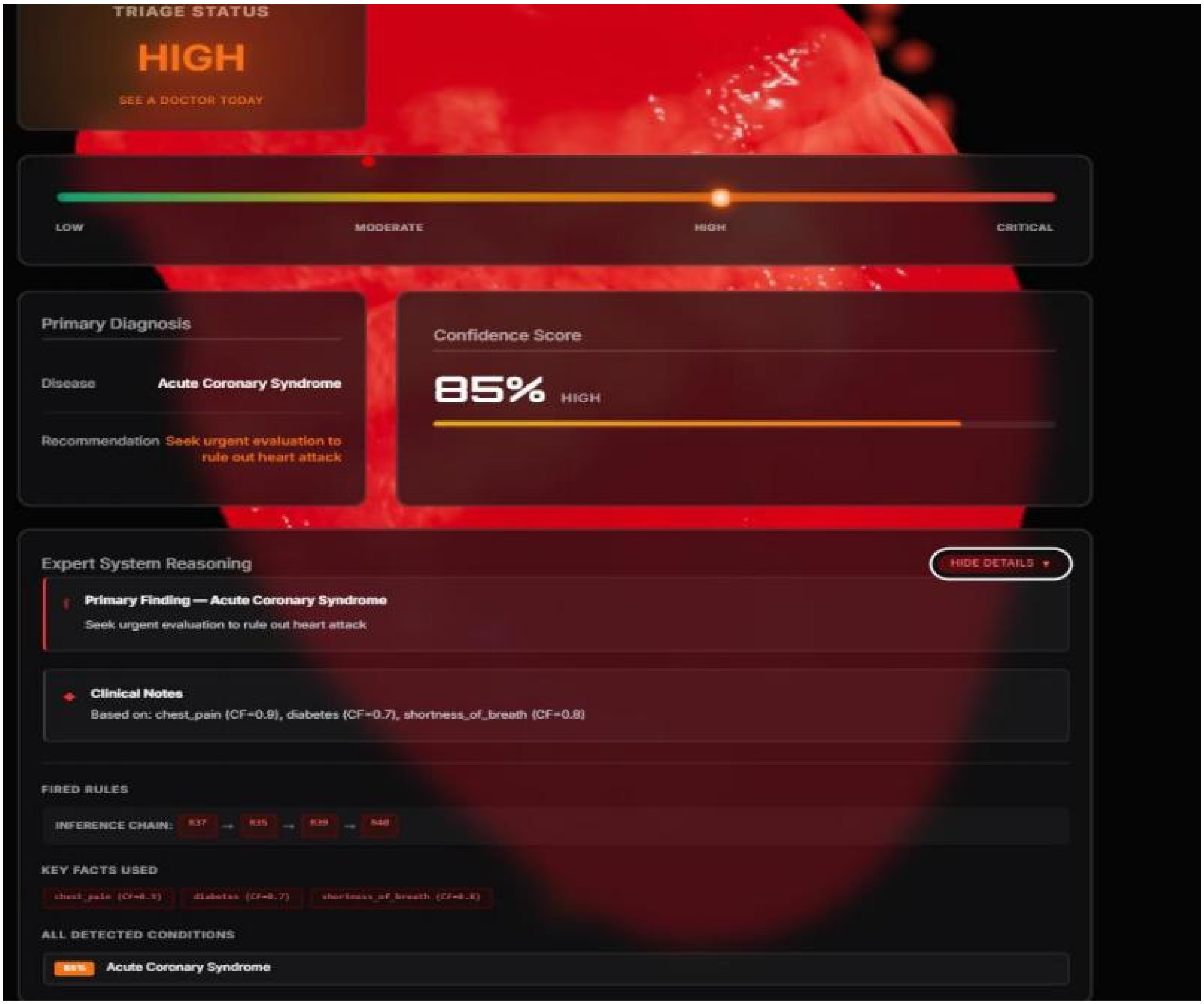
Cycle 5 (Finalizing Output)

- **Conflict Set:** {R39 (F5), R40 (F5)}
- **Resolution: R39 (F5)** wins (Urgency first).
- **Action:** Add **F7: Urgency = HIGH**.

Final System Output

Parameter	Result
Inferred Disease	Acute Coronary Syndrome (ACS)
Confidence (CF)	0.855
Urgency Level	HIGH
Medical Advice	Seek urgent evaluation to rule out heart attack

-The system output



Case 4: Low Urgency (Peripheral Edema)
-Manual solution

Initial Facts (Working Memory):

F1: Obesity (CF = 0.65)
F2: Edema (CF = 0.75) — Newest

Execution Trace:

Cycle 1
- **Conflict Set: {R12, R14, R15, R22}**
- **Resolution (LEX): Since the patient lacks respiratory symptoms (Shortness of breath) or Hypertension, the complex rules for Heart Failure fail to fire. The system identifies the isolated symptom of Edema.**
- **Action: Add F3: Disease = Peripheral Edema — Non-Cardiac Origin Likely.**

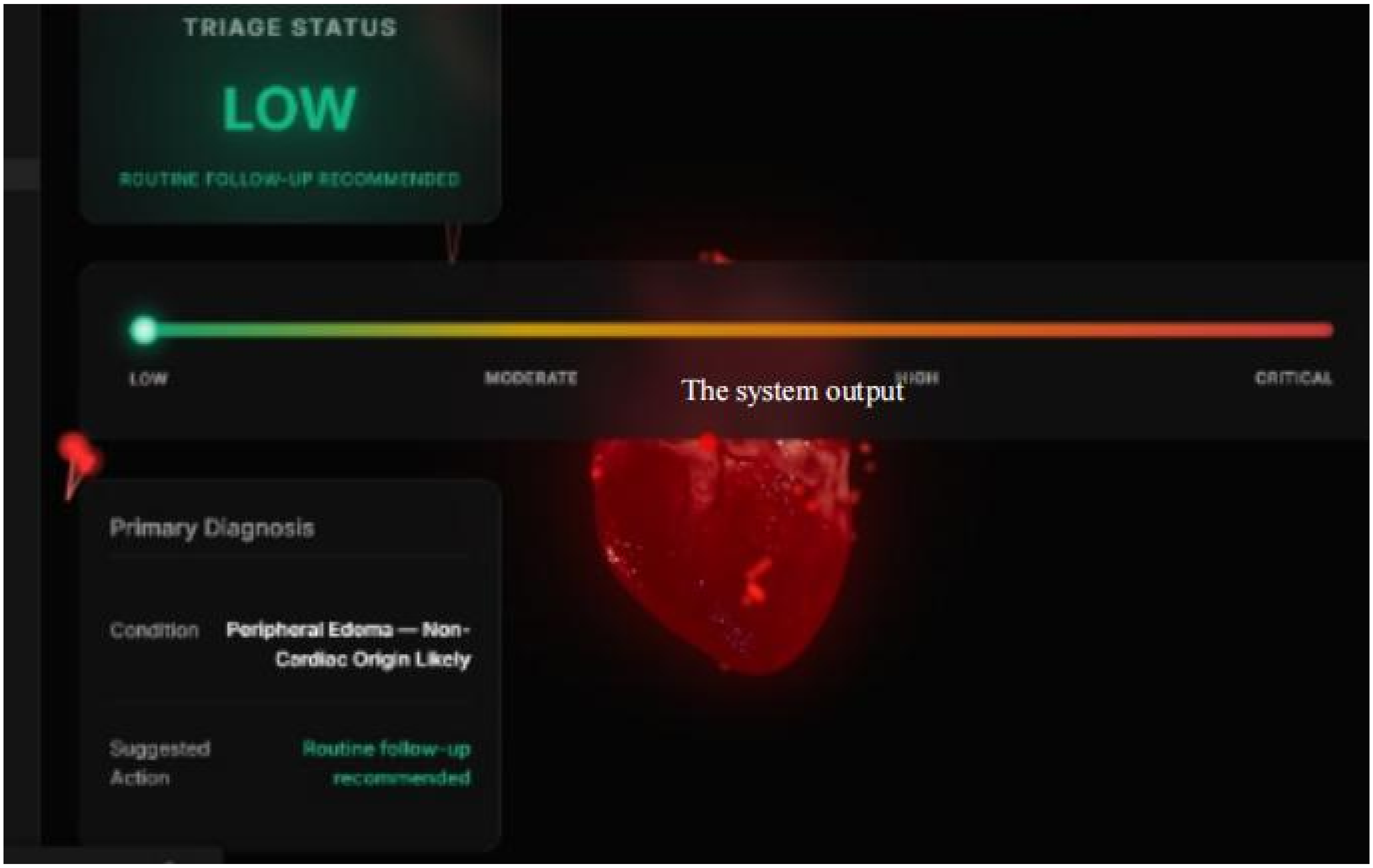
Cycle 2
Conflict Set: {Urgency Rule for Low}
Resolution (LEX): Stability of all vital signs triggers the lowest urgency level.
Action: Add F4: Urgency = LOW.

Cycle 3
- **Conflict Set: {Recommendation Rule}.**
- **Resolution (LEX): Selection of the appropriate advice for stable, non emergency cases.**
- **Action: Add F5: Recommendation exists.**

Cycle 4
- **Conflict Set: {R0}.**
- **Resolution (LEX): R0 fires as it matches the final recommendation fact.**
- **Action: Print final output**

Final Output:
Disease: Peripheral Edema — Non-Cardiac Origin Likely
Urgency: LOW
Advice: Routine follow-up recommended

The system output



Case 5: Critical Emergency (Acute Myocardial Infarction)

-Manual solution

Initial Facts (Working Memory):

F1: Chest Pain (CF = 0.90)

F2: Chest Tightness (CF = 0.75)

F3: Shortness of Breath (CF = 0.80) — Newest

Execution Trace:

Cycle 1

- **Conflict Set: {R25 (F1, F2, F3), R35 (F1, F3)}**
- **Resolution (LEX): R25 wins (Specificity: Matches 3 conditions compared to 2 in R35).**
- **Calculation: $Min(F1, F2, F3) = 0.75. CF_{R25} = 0.95 \times 0.75 = 0.7125.**
- **Action: Add F4: Disease = Acute Myocardial Infarction (MI) ($CF = 0.713).**

Cycle 2

- **Conflict Set: {R33 (F4), R34 (F4), R35 (F1, F3)}**
- **Resolution (LEX): R33 wins (Recency: Matches newest fact F4; Order: Appears before R34).**
- **Action: Add F5: Urgency = CRITICAL.**

Cycle 3

- **Conflict Set: {R34 (F4), R35 (F1, F3)}**
- **Resolution (LEX): R34 wins (Recency: Matches newest fact F4).**
- **Action: Add F6: Recommendation exists.**

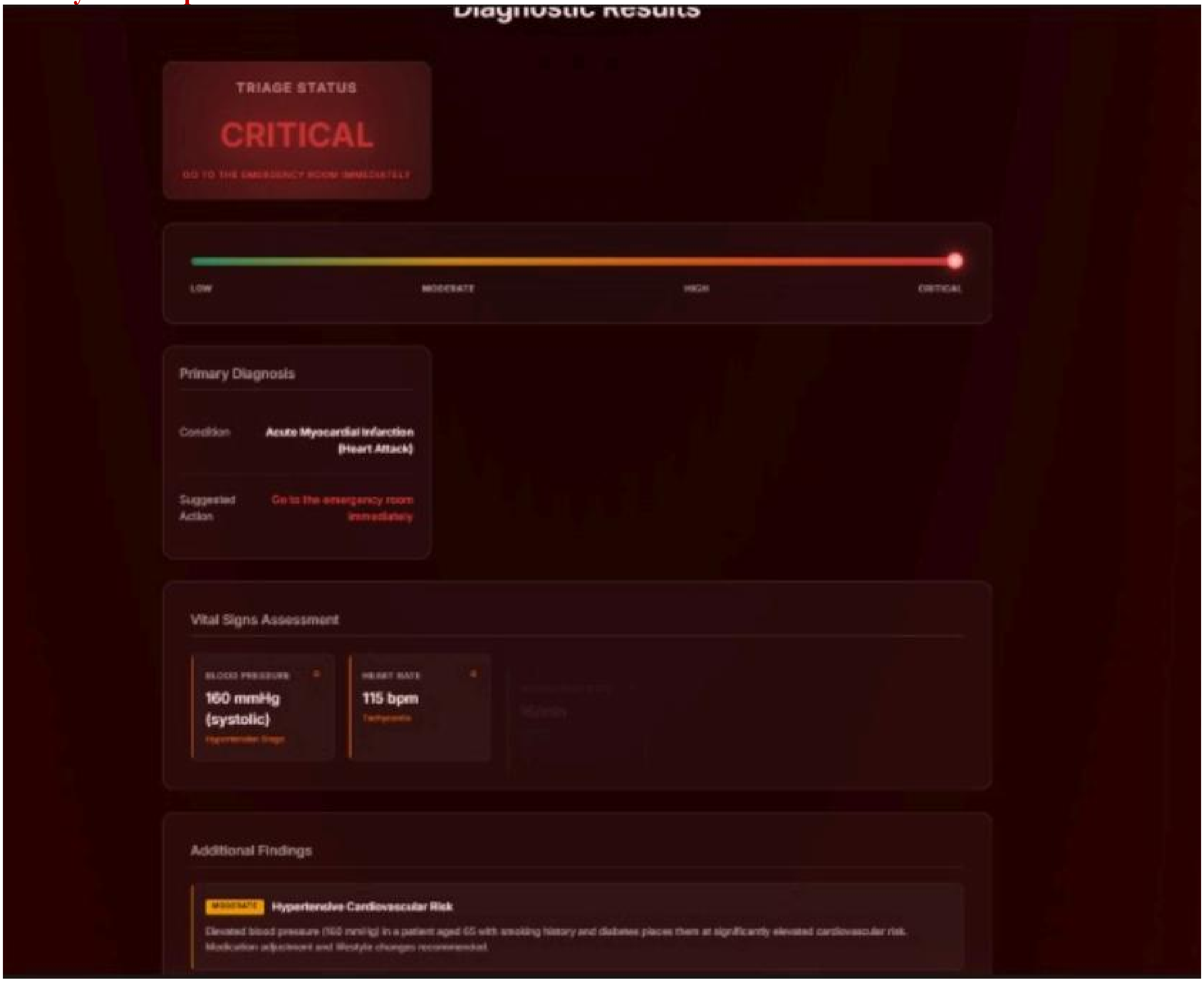
Cycle 4

- **Conflict Set: {R0 (F4, F5, F6), R35 (F1, F3)}**
- **Resolution (LEX): R0 fires to print the primary diagnosis.**
- **Action: Initial output generated for MI.**

Final Output:

Disease: Acute Myocardial Infarction (Heart Attack)
Urgency: CRITICAL
Advice: Go to the emergency room immediately and avoid physical activity

-The system output



Case 6: Moderate Urgency (Hypertensive Cardiovascular Risk)

-Manual solution

Initial Facts (Working Memory):

F1: Age > 65 (CF = 0.85$)
F2: Smoking History (CF = 0.70$)
F3: Blood Pressure = 160 (CF = 0.80$) — Newest

Execution Trace:

Cycle 1
- **Conflict Set: {R45, R46, R48}**
- **Resolution (LEX): R46 fires because it specifically addresses high blood pressure (Stage 2) in the context of cardiovascular risk factors (Age and Smoking).**
- **Calculation: Min(F1, F2, F3) = 0.70. CF_{R46} = 0.90* 0.70 = 0.63.**
- **Action: Add F4: Disease = Hypertensive Cardiovascular Risk.**

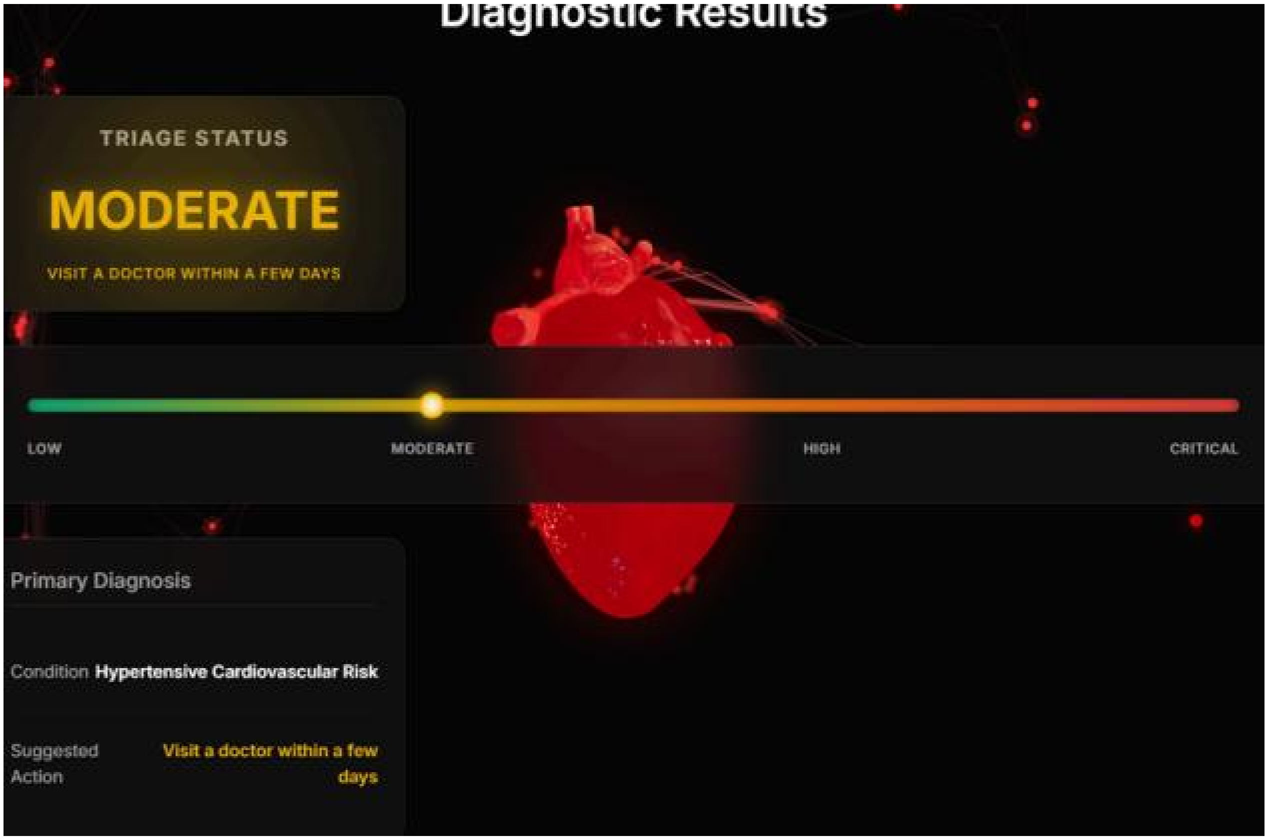
Cycle 2
- **Conflict Set: {Urgency Rule for Moderate}**
- **Resolution (LEX): Since the blood pressure is high (160) but there are no emergency symptoms like chest pain or syncope, the system assigns a moderate level.**
- **Action: Add F5: Urgency = MODERATE.**

Cycle 3
- **Conflict Set: {Recommendation Rule}**
- **Resolution (LEX): Selection of the advice for non-emergency hypertension management.**
- **Action: Add F6: Recommendation exists.**

Cycle 4
- **Conflict Set: {R0}**
- **Resolution (LEX): R0 fires to present the final diagnostic results.**
- **Action: Print final output.**

Final Output:
Disease: Hypertensive Cardiovascular Risk
Urgency: MODERATE
Advice: Visit a doctor within a few days

-The system output



3. Automated Testing (Pytest) Evaluation

The system's core inference engine was validated using the automated test suite in *test_diagnosis.py*. The system achieved a **100% success rate** across all functional test categories.

Test Case	Purpose & Evaluation Metric	Result
test_chf_diagnosis	Validates correctness of outputs for classic Chronic Heart Failure patterns.	**PASSED**
test_consistency	Ensures the system is deterministic; identical inputs yield identical CF levels.	**PASSED**
test_explanation_structure	Verifies transparency by checking for fired rules and key clinical facts.	**PASSED**
test_edge_cases	Ensures the engine handles null or empty inputs gracefully without crashing.	**PASSED**
test_insufficient_evidence	Validates safety thresholds by refusing to diagnose based on non-specific symptoms.	**PASSED**
test_disease_differentiation	Confirms conflict resolution logic priorities urgent MI over moderate AFib.	**PASSED**
test_chained_inference	Validates forward chaining (e.g., using inferred CHF to help diagnose ADHF).	**PASSED**
test_ahmed_hassan_scenario	Full-scale validation of the primary case study documented in the project.	**PASSED**

4. Detailed Scenario Evaluation: The "Ahmed Hassan" Case

To demonstrate the system's performance in a real healthcare context, we evaluated the case of Mr. Ahmed Hassan, a 67-year-old male with multiple risk factors.

Fact Acquisition: The system stored initial facts including shortness of breath, orthopnea, edema, and a background of heart disease.

Evidence Aggregation: The inference engine matched multiple rules (R1, R3). Because these rules were concordant, the system combined their Certainty Factors using the formula:

$$CF_{combined} = CF_{old} + CF_{new} - (CF_{old} \times CF_{new}).$$

Final Output: The cumulative evidence resulted in a final CF of **0.99** for **Acute Decompensated Heart Failure (ADHF)** with a **HIGH** urgency level.

5. System Limitations and Analysis

The evaluation phase highlighted specific boundaries within the current Knowledge Base:

Closed World Constraint: The system only identifies 10 predefined cardiac conditions; it cannot detect unknown diseases or non-cardiac issues.

Input Dependency: Performance is strictly determined by the quality and accuracy of user-provided data.

Threshold Rigidity: The system relies on fixed clinical measurements, which ensures high standards but eliminates the subjectivity a human specialist might apply.

Evidence Requirements: As proven in the "Safety Sensitivity" tests, the system will not fire rules if the minimum clinical pattern (e.g., a "triad" of symptoms) is not met.

6. Conclusion

The **CardKnowlogy** system successfully meets the requirements for a Telemedicine-based triage tool. It provides a transparent, evidence-based, and specialist-level assessment that is computationally efficient enough for rural or underserved areas. By prioritizing explainability and clinical safety, the system effectively guides users toward the necessary level of medical attention.